DSC 140A - Homework 1

due Monday, October 6 at 11:59PM

Problem 1. MLE Basics

Suppose we are given n data points drawn independently from a Normal distribution:

$$x_1, \ldots, x_n \overset{\text{iid}}{\sim} \text{Normal}(\mu, \sigma^2)$$

a) What is the likelihood function $L(\mu, \sigma^2 | x_1, \ldots, x_n)$?

Solution:

$$\begin{aligned} L(\mu, \sigma^2 | x_1, \ldots, x_n) &= \prod_{i=1}^{n} f(x_i | \mu, \sigma^2) \\ &= \prod_{i=1}^{n} \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x_i - \mu)^2}{2\sigma^2}\right) \\ &= \left(\frac{1}{\sqrt{2\pi\sigma^2}}\right)^n \exp\left(-\frac{1}{2\sigma^2}\sum_{i=1}^{n}(x_i - \mu)^2\right) \end{aligned}$$

b) What is the maximum likelihood estimator $\hat{\mu}_{MLE}(x_1, \ldots, x_n)$?

Solution:

$$\begin{aligned} \hat{\mu}_{MLE} &= \arg\max_{\mu} L(\mu, \sigma^2 | x_1, \ldots, x_n) \\ &= \arg\max_{\mu} \left[\left(\frac{1}{\sqrt{2\pi\sigma^2}}\right)^n \exp\left(-\frac{1}{2\sigma^2}\sum_{i=1}^{n}(x_i - \mu)^2\right)\right] \\ &= \arg\max_{\mu} \left[-\frac{1}{2\sigma^2}\sum_{i=1}^{n}(x_i - \mu)^2\right] \end{aligned}$$

Since this is a negative sum of squares, the maximum is achieved when

$$\begin{aligned} \frac{\partial}{\partial\mu}\frac{1}{2\sigma^2}\sum_{i=1}^{n}(x_i - \hat{\mu}_{MLE})^2 &= 0 \\ \sum_{i=1}^{n} x_i - n\hat{\mu}_{MLE} &= 0 \\ \hat{\mu}_{MLE} &= \frac{1}{n}\sum_{i=1}^{n} x_i \\ \implies \hat{\mu}_{MLE} &= \bar{x} \end{aligned}$$

c) What is the expected value of the estimator $\mathbb{E}(\hat{\mu}_{MLE})$? Is it biased?

Solution:

$$\mathbb{E}(\hat{\mu}_{MLE}) = \mathbb{E}\left(\frac{1}{n}\sum_{i=1}^{n} x_i\right) = \frac{1}{n}\sum_{i=1}^{n}\mathbb{E}(x_i) = \frac{1}{n}\sum_{i=1}^{n}\mu = \mu$$

Since $\mathbb{E}(\hat{\mu}_{MLE}) = \mu$, $\hat{\mu}_{MLE}$ is unbiased.

d) What is the variance of estimator $\mathrm{Var}(\hat{\mu}_{MLE})$? Is it consistent?

Solution:

$$\mathrm{Var}(\hat{\mu}_{MLE}) = \mathrm{Var}\left(\frac{1}{n}\sum_{i=1}^{n} x_i\right) = \frac{1}{n^2}\sum_{i=1}^{n}\mathrm{Var}(x_i) = \frac{1}{n^2}\sum_{i=1}^{n}\sigma^2 = \frac{\sigma^2}{n}$$

Since $\lim_{n\to\infty}\mathrm{Var}(\hat{\mu}_{MLE}) = 0$, $\hat{\mu}_{MLE}$ is consistent.

e) What is the maximum likelihood estimator $\hat{\sigma}_{MLE}$?

Solution:

$$\begin{aligned}
\hat{\sigma}_{MLE} &= \arg\max_{\sigma} L(\mu, \sigma | x_1, \ldots, x_n) \\
&= \arg\max_{\sigma}\left[\left(\frac{1}{\sqrt{2\pi\sigma^2}}\right)^n \exp\left(-\frac{1}{2\sigma^2}\sum_{i=1}^{n}(x_i-\mu)^2\right)\right] \\
&= \arg\min_{\sigma}\left[\frac{n}{2}\ln(2\pi) + n\ln(\sigma) + \frac{1}{2\sigma^2}\sum_{i=1}^{n}(x_i-\mu)^2\right] \\
0 &= \frac{\partial}{\partial\sigma}\left[\frac{n}{2}\ln(2\pi) + n\ln(\sigma) + \frac{1}{2\sigma^2}\sum_{i=1}^{n}(x_i-\mu)^2\right] \\
0 &= \frac{n}{\hat{\sigma}_{MLE}} - \frac{1}{\hat{\sigma}^3_{MLE}}\sum_{i=1}^{n}(x_i-\mu)^2 \\
n\hat{\sigma}^2_{MLE} &= \sum_{i=1}^{n}(x_i-\mu)^2 \\
\implies \hat{\sigma}_{MLE} &= \sqrt{\frac{1}{n}\sum_{i=1}^{n}(x_i-\mu)^2}
\end{aligned}$$

Problem 2. MLE Basics - Matrix Version

Suppose we are given n data points drawn independently from a d-dimensional Normal distribution with mean $\mu \in \mathbb{R}^d$ and covariance matrix $\Sigma \in \mathbb{R}^{d \times d}$:

$$x_1, \ldots, x_n \overset{\text{iid}}{\sim} \text{Normal}(\mu, \Sigma)$$

a) What is the likelihood function $L(\mu, \Sigma | x_1, \ldots, x_n)$?

Solution:

$$\begin{aligned} L(\mu, \Sigma | x_1, \ldots, x_n) &= \prod_{i=1}^{n} f(x_i | \mu, \Sigma) \\ &= \prod_{i=1}^{n} \frac{1}{\sqrt{(2\pi)^d |\Sigma|}} \exp\left(-\frac{1}{2} (x_i - \mu)^T \Sigma^{-1} (x_i - \mu) \right) \\ &= \left(\frac{1}{\sqrt{(2\pi)^d |\Sigma|}} \right)^n \exp\left(-\frac{1}{2} \sum_{i=1}^{n} (x_i - \mu)^T \Sigma^{-1} (x_i - \mu) \right) \end{aligned}$$

b) What is the maximum likelihood estimate $\hat{\mu}_{MLE}$?

Solution:

$$\begin{aligned} \hat{\mu}_{MLE} &= \arg\max_{\mu} L(\mu, \Sigma | x_1, \ldots, x_n) \\ &= \arg\max_{\mu} \left[\left(\frac{1}{\sqrt{(2\pi)^d |\Sigma|}} \right)^n \exp\left(-\frac{1}{2} \sum_{i=1}^{n} (x_i - \mu)^T \Sigma^{-1} (x_i - \mu) \right) \right] \\ &= \arg\max_{\mu} \left[-\frac{1}{2} \sum_{i=1}^{n} (x_i - \mu)^T \Sigma^{-1} (x_i - \mu) \right] \end{aligned}$$

Since this is a negative sum of quadratic terms, the maximum is achieved when

$$\begin{aligned} \frac{\partial}{\partial \mu} \left[-\frac{1}{2} \sum_{i=1}^{n} (x_i - \mu)^T \Sigma^{-1} (x_i - \mu) \right] &= 0 \\ \sum_{i=1}^{n} \Sigma^{-1} (x_i - \mu) &= 0 \\ \implies \hat{\mu}_{MLE} &= \frac{1}{n} \sum_{i=1}^{n} x_i \\ \hat{\mu}_{MLE} &= \bar{x} \end{aligned}$$

c) What is the expected value of the estimator $\mathbb{E}(\hat{\mu}_{MLE})$? Is it biased?

Solution:

$$\mathbb{E}(\hat{\mu}_{MLE}) = \mathbb{E}\left(\frac{1}{n} \sum_{i=1}^{n} x_i \right) = \frac{1}{n} \sum_{i=1}^{n} \mathbb{E}(x_i) = \frac{1}{n} \sum_{i=1}^{n} \mu = \mu$$

Since $\mathbb{E}(\hat{\mu}_{MLE}) = \mu$, $\hat{\mu}_{MLE}$ is unbiased.

d) What is the variance of the estimator $\mathrm{Var}(\hat{\mu}_{MLE})$? Is it consistent?

Solution:

$$\mathrm{Var}(\hat{\mu}_{MLE}) = \mathrm{Var}\left(\frac{1}{n}\sum_{i=1}^{n} x_i\right) = \frac{1}{n^2}\sum_{i=1}^{n}\mathrm{Var}(x_i) = \frac{1}{n^2}\sum_{i=1}^{n}\Sigma = \frac{\Sigma}{n}$$

Since $\lim_{n\to\infty}\mathrm{Var}(\hat{\mu}_{MLE}) = 0$, $\hat{\mu}_{MLE}$ is consistent.

e) (Optional, not graded) What is the maximum likelihood estimator $\hat{\Sigma}_{MLE}$?

Solution:

$$\begin{aligned}
\hat{\Sigma}_{MLE} &= \arg\max_{\Sigma} L(\mu, \Sigma | x_1, \ldots, x_n) \\
&= \arg\max_{\Sigma}\left[\left(\frac{1}{\sqrt{(2\pi)^d|\Sigma|}}\right)^n \exp\left(-\frac{1}{2}\sum_{i=1}^{n}(x_i-\mu)^T\Sigma^{-1}(x_i-\mu)\right)\right] \\
&= \arg\min_{\Sigma}\left[\frac{n}{2}\ln(2\pi) + n\ln(|\Sigma|) + \frac{1}{2}\sum_{i=1}^{n}(x_i-\mu)^T\Sigma^{-1}(x_i-\mu)\right] \\
0 &= \frac{\partial}{\partial\Sigma}\left[\frac{n}{2}\ln(2\pi) + n\ln(|\Sigma|) + \frac{1}{2}\sum_{i=1}^{n}(x_i-\mu)^T\Sigma^{-1}(x_i-\mu)\right] \\
0 &= \frac{n}{2|\hat{\Sigma}|}\frac{\partial}{\partial\Sigma}|\hat{\Sigma}| - \frac{1}{2}\left[\sum_{i=1}^{n}(x_i-\mu)(x_i-\mu)^T\right]\hat{\Sigma}^{-1}\frac{\partial}{\partial\Sigma}\hat{\Sigma}\hat{\Sigma}^{-1} \\
0 &= \frac{n}{2}\hat{\Sigma}^{-1} - \frac{1}{2}\left[\sum_{i=1}^{n}(x_i-\mu)(x_i-\mu)^T\right]\hat{\Sigma}^{-1}\hat{\Sigma}^{-1} \\
\implies \hat{\Sigma}_{MLE} &= \frac{1}{n}\sum_{i=1}^{n}(x_i-\mu)(x_i-\mu)^T
\end{aligned}$$

Problem 3. MLE Basics - Exponential Distribution

Suppose we are given n data points drawn independently from an Exponential distribution:

$$x_1, \ldots, x_n \overset{\text{iid}}{\sim} \text{Exponential}(\theta)$$

The probability density function for a single point x_i is described as:

$$f(x_i, \theta) = \begin{cases} \theta \exp(-\theta x_i) & x_i \geq 0 \\ 0 & x_i < 0 \end{cases} \tag{1}$$

a) Describe a real-world outcome that we could model using this distribution.

Solution: This distribution can model things like radioactive decay where θ is related to the half-life of the particle.

b) Derive an expression for $\hat{\theta}_{MLE}$, the maximum likelihood estimator of θ?

Solution:

$$L(\theta | x_1, \ldots, x_n) = \prod_{i=1}^{n} f(x_i|\theta) = \begin{cases} \theta^n \exp\left(-\theta \sum_{i=1}^{n} x_i\right) & \forall x_i \geq 0 \\ 0 & \exists x_i < 0 \end{cases}$$

$$\begin{aligned} \hat{\theta}_{MLE} &= \arg\max_{\theta} \left[\theta^n \exp\left(-\theta \sum_{i=1}^{n} x_i \right) \right] \\ &= \arg\max_{\theta} \left[n \ln(\theta) - \theta \sum_{i=1}^{n} x_i \right] \\ 0 &= \frac{\partial}{\partial \theta} \left[n \ln(\theta) - \theta \sum_{i=1}^{n} x_i \right] \\ 0 &= \frac{n}{\theta} - \sum_{i=1}^{n} x_i \\ \implies \hat{\theta}_{MLE} &= \frac{n}{\sum_{i=1}^{n} x_i} \end{aligned}$$

c) Describe how you could compute $\hat{\theta}_{MLE}$ using only a summary statistic of the data?

Solution: I could use the sample mean of the data points, $\bar{x}$, to estimate $\hat{\theta}_{MLE}$ since $\hat{\theta}_{MLE} = \bar{x}^{-1}$.

d) Say you compute the MLE estimator using n data points, and then delete the data. Describe how you can update the MLE estimator with another point x_{n+1}? The final equation must be in terms of the summary statistic computed in last part, x_{n+1}, and n.

Solution: You can use the sample mean of the data points, $\bar{x}$, to estimate the new $\hat{\theta}_{MLE}$ as

$$\hat{\theta}_{MLE} = \frac{n+1}{n\bar{x} + x_{n+1}}$$

Problem 4. MLE without First-Derivatives

Suppose we have 5 data points drawn independently from a **Uniform**(a, b) distribution:

$$1.2, 2.1, 1.3, 10.5, 5.0$$

a) What are the maximum likelihood estimates for the parameters a and b?

Solution:

$$L(a, b|x_1, \ldots, x_n) = \prod_{i=1}^{n} f(x_i|a, b) = \begin{cases} \left(\frac{1}{b-a}\right)^n & \forall x_i \in [a, b] \\ 0 & \exists x_i \notin [a, b] \end{cases}$$

To maximize this, you need to find $\arg\min_{a,b}(b - a)$ such that $\forall x_i, x_i \in [a, b]$. This is achieved when $\hat{a} = \min(x_1, \ldots, x_n)$ and $\hat{b} = \max(x_1, \ldots, x_n)$. In this case, the MLE estimates are $\hat{a} = 1.2$ and $\hat{b} = 10.5$.

b) Suppose that we have n data points $x_1, \ldots, x_n$ drawn independently from **Uniform**(a, b) distribution where $a < b$. What are the maximum likelihood estimators for a and b?

Solution: Using the same logic as above, the MLE estimates are $\hat{a} = \min(x_i)$ and $\hat{b} = \max(x_i)$.

Problem 5. Simple Linear Regression via MLE

In this problem, we'll use maximum likelihood estimation to fit the "best fit line" between x and y from a dataset of n points $(x_1, y_1), \ldots, (x_n, y_n)$.

We assume that each y_i is function of x_i observed with measurement error $\epsilon_i \overset{\text{iid}}{\sim} \text{Normal}(0, \sigma^2)$. Formally, we can write:

$$y_i|x_i \sim ax_i + b + \epsilon_i \tag{2}$$

a) For general data $(x_1, y_1), \ldots, (x_n, y_n)$, write down the expression for the likelihood and the log-likelihood functions as functions of a, b, and σ.

Solution:

$$\begin{aligned} L(a, b, \sigma|x_1, \ldots, x_n, y_1, \ldots, y_n) = \prod_{i=1}^{n} f(y_i|x_i, a, b, \sigma) &= \prod_{i=1}^{n} \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(y_i - (ax_i + b))^2}{2\sigma^2}\right) \\ &= \left(\frac{1}{\sqrt{2\pi\sigma^2}}\right)^n \exp\left(-\frac{1}{2\sigma^2}\sum_{i=1}^{n}(y_i - (ax_i + b))^2\right) \end{aligned}$$

$$\begin{aligned} \ell(a, b, \sigma|x_1, \ldots, x_n, y_1, \ldots, y_n) &= \sum_{i=1}^{n} \ln\left(\frac{1}{\sqrt{2\pi\sigma^2}}\right) - \frac{(y_i - (ax_i + b))^2}{2\sigma^2} \\ &= -\frac{n}{2}\ln(2\pi) - n\ln(\sigma) - \frac{1}{2\sigma^2}\sum_{i=1}^{n}(y_i - (ax_i + b))^2 \end{aligned}$$

b) Derive the maximum likelihood estimates $\hat{a}_{MLE}$ and $\hat{b}_{MLE}$.

Solution:

$$\arg\max_{a,b}\left[\prod_{i=1}^{n}\frac{1}{\sqrt{2\pi\sigma^2}}\exp\left(-\frac{(y_i-(ax_i+b))^2}{2\sigma^2}\right)\right]=\arg\max_{a,b}\left[-\sum_{i=1}^{n}\frac{(y_i-(ax_i+b))^2}{2\sigma^2}\right]$$

$$\begin{aligned}
0&=\frac{\partial}{\partial b}\left[-\sum_{i=1}^{n}\frac{(y_i-(ax_i+b))^2}{2\sigma^2}\right]\\
0&=\sum_{i=1}^{n}(y_i-(\hat{a}x_i+\hat{b}))\\
0&=\sum_{i=1}^{n}y_i-\hat{a}\sum_{i=1}^{n}x_i-\hat{b}n\\
\implies \hat{b}_{MLE}&=\frac{1}{n}\sum_{i=1}^{n}y_i-\hat{a}\frac{1}{n}\sum_{i=1}^{n}x_i=\bar{y}-\hat{a}\bar{x}\\
0&=\frac{\partial}{\partial a}\left[-\sum_{i=1}^{n}\frac{(y_i-(ax_i+b))^2}{2\sigma^2}\right]\\
0&=\sum_{i=1}^{n}(y_i-(\hat{a}x_i+\hat{b}))x_i\\
0&=\sum_{i=1}^{n}y_ix_i-\hat{a}\sum_{i=1}^{n}x_i^2-\hat{b}\sum_{i=1}^{n}x_i\\
0&=\sum_{i=1}^{n}y_ix_i-\hat{a}\sum_{i=1}^{n}x_i^2-\left(\frac{1}{n}\sum_{i=1}^{n}y_i-\hat{a}\frac{1}{n}\sum_{i=1}^{n}x_i\right)\sum_{i=1}^{n}x_i\\
\implies \hat{a}_{MLE}&=\frac{n\sum_{i=1}^{n}y_ix_i-\sum_{i=1}^{n}y_i\sum_{i=1}^{n}x_i}{n\sum_{i=1}^{n}x_i^2-\sum_{i=1}^{n}x_i\sum_{i=1}^{n}x_i}
\end{aligned}$$

We will now consider a dataset generated from the model:

$$y|x \sim \text{Normal}(w_0 + w_1 x, \sigma^2) \tag{3}$$

We assume that the values of σ is known and use the dataset to estimate the values of w_0 and w_1.

c) Show that the given model in (2) and (3) are equivalent.

Solution: The models are equivalent because the likelihood functions are the same. The likelihood function for the model in (2) is:

$$L(a, b, \sigma|x_1, \ldots, x_n, y_1, \ldots, y_n) = \prod_{i=1}^{n} \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(y_i - (ax_i + b))^2}{2\sigma^2}\right)$$

The likelihood function for the model in (3) is:

$$L(w_0, w_1, \sigma|x_1, \ldots, x_n, y_1, \ldots, y_n) = \prod_{i=1}^{n} \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(y_i - (w_0 + w_1 x_i))^2}{2\sigma^2}\right)$$

These are the same if $w_0 = b$ and $w_1 = a$.

We estimate the values of w_0 and w_1 using the dataset and get $\hat{w_0}$ and $\hat{w_1}$ respectively. We can then predict the outcome using a *fitted model* of the form:

$$\hat{f}(x_i) = \hat{y_i} = \hat{w_0} + \hat{w_1} x_i$$

Given a fitted model, we can measure the *residual sum of squares* as

$$\text{RSS}(\hat{f}) = \sum_{i=1}^{n} (y_i - \hat{f}(x_i))^2 = \sum_{i=1}^{n} (y_i - \hat{w_0} - \hat{w_1} x_i)^2 \tag{4}$$

d) Show that the RSS is minimized by a fitted model with parameters:

$$\hat{w_0} = \bar{y} - \hat{w_1}\bar{x}$$

$$\hat{w_1} = \frac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^{n}(x_i - \bar{x})^2}$$

Here, $\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i$ and $\bar{y} = \frac{1}{n}\sum_{i=1}^{n} y_i$ denote the mean values of x_i and y_i in the dataset.

Solution: Since the RSS is a sum of squares, it is minimized when the gradient is 0.

$$\begin{aligned}
0 &= \frac{\partial}{\partial w_0}\left[\sum_{i=1}^{n}(y_i - (w_0 + w_1 x_i))^2\right] \\
0 &= \frac{\partial}{\partial w_0}\left[\sum_{i=1}^{n}(y_i - (\bar{y} - \hat{w_1}\bar{x} + w_1 x_i))^2\right] \\
0 &= \sum_{i=1}^{n}(y_i - \bar{y} + \hat{w_1}\bar{x} - \hat{w_1} x_i) \\
0 &= \sum_{i=1}^{n} y_i - \bar{y}n + \hat{w_1}\bar{x}n - \hat{w_1}\sum_{i=1}^{n} x_i \\
0 &= \sum_{i=1}^{n} y_i - \sum_{i=1}^{n} y_i + \hat{w_1}\sum_{i=1}^{n} x_i - \hat{w_1}\sum_{i=1}^{n} x_i = 0 \\
\implies \hat{w_0} &= \bar{y} - \hat{w_1}\bar{x} \\
0 &= \frac{\partial}{\partial w_1}\left[\sum_{i=1}^{n}(y_i - (\bar{y} - \hat{w_1}\bar{x} + w_1 x_i))^2\right] \\
0 &= \sum_{i=1}^{n}(y_i - \bar{y} + \hat{w_1}(\bar{x} - x_i))(x_i - \bar{x}) \\
0 &= \sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y}) - \hat{w_1}\sum_{i=1}^{n}(x_i - \bar{x})^2 = 0 \\
\implies \hat{w_1} &= \frac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^{n}(x_i - \bar{x})^2}
\end{aligned}$$

e) Show that $\text{Var}(\bar{y}) = \frac{\sigma^2}{n}$. *Note: you can treat the values of $x_1, \ldots, x_n$ as fixed in your derivation.*

Solution: Since $y_i \sim \text{Normal}(w_0 + w_1 x_i, \sigma^2)$, $\text{Var}(y_i) = \sigma^2$.

$$\text{Var}(\bar{y}) = \text{Var}\left(\frac{1}{n}\sum_{i=1}^{n} y_i\right) = \frac{1}{n^2}\sum_{i=1}^{n}\text{Var}(y_i) = \frac{1}{n^2}\sum_{i=1}^{n}\sigma^2 = \frac{\sigma^2}{n}$$

f) Show that the simple linear regression line will always pass through the point $(\bar{x}, \bar{y})$.

Solution: The equation of the line is $\bar{y} = \hat{w_0} + \hat{w_1}\bar{x}$. Plugging in $\hat{w_0} = \bar{y} - \hat{w_1}\bar{x}$, we get $\bar{y} = \bar{y} - \hat{w_1}\bar{x} + \hat{w_1}\bar{x}$ which becomes $\bar{y} = \bar{y}$ which is true.

We now consider a simpler model where $w_0 = 0$. In this case, we assume that each data point is generated from the model $y|x \sim \text{Normal}(w_1 x, \sigma^2)$, and the fitted model has the form $\hat{f}(x_i) = \hat{w} x_i$.

g) Show that the maximum likelihood estimate can be expressed as

$$\hat{w_1} = \frac{\sum_{i=1}^n x_i y_i}{\sum_{i=1}^n x_i^2}$$

Solution:

$$L(w_1, \sigma | x_1, \ldots, x_n, y_1, \ldots, y_n) = \prod_{i=1}^n \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(y_i - w_1 x_i)^2}{2\sigma^2}\right)$$

$$\arg\max_{w_1,\sigma} \left[\prod_{i=1}^n \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(y_i - w_1 x_i)^2}{2\sigma^2}\right)\right] = \arg\max_{w_1,\sigma} \left[-\sum_{i=1}^n \frac{(y_i - w_1 x_i)^2}{2\sigma^2}\right]$$

$$0 = \frac{\partial}{\partial w_1}\left[-\sum_{i=1}^n \frac{(y_i - w_1 x_i)^2}{2\sigma^2}\right]$$

$$0 = \sum_{i=1}^n (y_i - w_1 x_i) x_i$$

$$0 = \sum_{i=1}^n y_i x_i - w_1 \sum_{i=1}^n x_i^2$$

$$\implies \hat{w_1} = \frac{\sum_{i=1}^n x_i y_i}{\sum_{i=1}^n x_i^2}$$

h) Show that the maximum likelihood estimator is (related to) the correlation between x and y.

Solution: The correlation between x and y is given by

$$\rho = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^n (x_i - \bar{x})^2 \sum_{i=1}^n (y_i - \bar{y})^2}}$$

Since the line goes through the origin, we treat it as if the data is centered at the origin, i.e. $\bar{x} = 0$ and $\bar{y} = 0$. Thus,

$$\rho = \frac{\sum_{i=1}^n x_i y_i}{\sqrt{\sum_{i=1}^n x_i^2 \sum_{i=1}^n y_i^2}}$$

$$\hat{w_1} = \rho \sqrt{\frac{\sum_{i=1}^n y_i^2}{\sum_{i=1}^n x_i^2}}$$

$$= \rho \sqrt{\frac{\text{Var}(y)}{\text{Var}(x)}}$$

$$= \rho \frac{\sigma_y}{\sigma_x}$$

i) Show that the best-fit line in this case will pass through the origin.

Solution:

$$\hat{f}(0) = \hat{w_1} \cdot 0 = 0$$

Hence, the line passes through the origin.

j) Describe a task where we would be better off using the simpler model.

Solution: We would be better off using the simpler model if the relationship between x and y is linear. That is, if you know that when $x = 0$, $y = 0$. One such example is in the case of force and acceleration.

Problem 6. Complete the Google form at the following link.

Responding to this form is mandatory as it is worth 1 point. Please note that it's not the content that is graded but whether you filled the form or not.